

$x_0$  to  $\sigma_i(v_0)$  and let  $\sigma_0$  be the singular 0-simplex with image  $x_0$ . We can view  $\tau_i$  as a singular 1-simplex, a map  $\tau_i: [v_0, v_1] \rightarrow X$ , and then we have  $\partial\tau_i = \sigma_i - \sigma_0$ . Hence  $\partial(\sum_i n_i \tau_i) = \sum_i n_i \sigma_i - \sum_i n_i \sigma_0 = \sum_i n_i \sigma_i$  since  $\sum_i n_i = 0$ . Thus  $\sum_i n_i \sigma_i$  is a boundary, which shows that  $\text{Ker } \varepsilon \subset \text{Im } \partial_1$ .  $\square$

**Proposition 2.8.** *If  $X$  is a point, then  $H_n(X) = 0$  for  $n > 0$  and  $H_0(X) \approx \mathbb{Z}$ .*

**Proof:** In this case there is a unique singular  $n$ -simplex  $\sigma_n$  for each  $n$ , and  $\partial(\sigma_n) = \sum_i (-1)^i \sigma_{n-1}$ , a sum of  $n+1$  terms, which is therefore 0 for  $n$  odd and  $\sigma_{n-1}$  for  $n$  even,  $n \neq 0$ . Thus we have the chain complex

$$\cdots \rightarrow \mathbb{Z} \xrightarrow{\approx} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\approx} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0$$

with boundary maps alternately isomorphisms and trivial maps, except at the last  $\mathbb{Z}$ . The homology groups of this complex are trivial except for  $H_0 \approx \mathbb{Z}$ .  $\square$

It is often very convenient to have a slightly modified version of homology for which a point has trivial homology groups in all dimensions, including zero. This is done by defining the **reduced homology groups**  $\tilde{H}_n(X)$  to be the homology groups of the augmented chain complex

$$\cdots \rightarrow C_2(X) \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0$$

where  $\varepsilon(\sum_i n_i \sigma_i) = \sum_i n_i$  as in the proof of Proposition 2.7. Here we had better require  $X$  to be nonempty, to avoid having a nontrivial homology group in dimension  $-1$ . Since  $\varepsilon\partial_1 = 0$ ,  $\varepsilon$  vanishes on  $\text{Im } \partial_1$  and hence induces a map  $H_0(X) \rightarrow \mathbb{Z}$  with kernel  $\tilde{H}_0(X)$ , so  $H_0(X) \approx \tilde{H}_0(X) \oplus \mathbb{Z}$ . Obviously  $H_n(X) \approx \tilde{H}_n(X)$  for  $n > 0$ .

Formally, one can think of the extra  $\mathbb{Z}$  in the augmented chain complex as generated by the unique map  $[\emptyset] \rightarrow X$  where  $[\emptyset]$  is the empty simplex, with no vertices. The augmentation map  $\varepsilon$  is then the usual boundary map since  $\partial[v_0] = [\hat{v}_0] = [\emptyset]$ .

Readers who know about the fundamental group  $\pi_1(X)$  may wish to make a detour here to look at §2.A where it is shown that  $H_1(X)$  is the abelianization of  $\pi_1(X)$  whenever  $X$  is path-connected. This result will not be needed elsewhere in the chapter, however.

## Homotopy Invariance

The first substantial result we will prove about singular homology is that homotopy equivalent spaces have isomorphic homology groups. This will be done by showing that a map  $f: X \rightarrow Y$  induces a homomorphism  $f_*: H_n(X) \rightarrow H_n(Y)$  for each  $n$ , and that  $f_*$  is an isomorphism if  $f$  is a homotopy equivalence.

For a map  $f: X \rightarrow Y$ , an induced homomorphism  $f_\# : C_n(X) \rightarrow C_n(Y)$  is defined by composing each singular  $n$ -simplex  $\sigma: \Delta^n \rightarrow X$  with  $f$  to get a singular  $n$ -simplex

$f_{\#}(\sigma) = f\sigma : \Delta^n \rightarrow Y$ , then extending  $f_{\#}$  linearly via  $f_{\#}(\sum_i n_i \sigma_i) = \sum_i n_i f_{\#}(\sigma_i) = \sum_i n_i f\sigma_i$ . The maps  $f_{\#} : C_n(X) \rightarrow C_n(Y)$  satisfy  $f_{\#}\partial = \partial f_{\#}$  since

$$\begin{aligned} f_{\#}\partial(\sigma) &= f_{\#}(\sum_i (-1)^i \sigma | [v_0, \dots, \hat{v}_i, \dots, v_n]) \\ &= \sum_i (-1)^i f\sigma | [v_0, \dots, \hat{v}_i, \dots, v_n] = \partial f_{\#}(\sigma) \end{aligned}$$

Thus we have a diagram

$$\begin{array}{ccccccc} \dots & \longrightarrow & C_{n+1}(X) & \xrightarrow{\partial} & C_n(X) & \xrightarrow{\partial} & C_{n-1}(X) \longrightarrow \dots \\ & & \downarrow f_{\#} & & \downarrow f_{\#} & & \downarrow f_{\#} \\ \dots & \longrightarrow & C_{n+1}(Y) & \xrightarrow{\partial} & C_n(Y) & \xrightarrow{\partial} & C_{n-1}(Y) \longrightarrow \dots \end{array}$$

such that in each square the composition  $f_{\#}\partial$  equals the composition  $\partial f_{\#}$ . A diagram of maps with the property that any two compositions of maps starting at one point in the diagram and ending at another are equal is called a **commutative diagram**. In the present case commutativity of the diagram is equivalent to the commutativity relation  $f_{\#}\partial = \partial f_{\#}$ , but commutative diagrams can contain commutative triangles, pentagons, etc., as well as commutative squares.

The fact that the maps  $f_{\#} : C_n(X) \rightarrow C_n(Y)$  satisfy  $f_{\#}\partial = \partial f_{\#}$  is also expressed by saying that the  $f_{\#}$ 's define a **chain map** from the singular chain complex of  $X$  to that of  $Y$ . The relation  $f_{\#}\partial = \partial f_{\#}$  implies that  $f_{\#}$  takes cycles to cycles since  $\partial\alpha = 0$  implies  $\partial(f_{\#}\alpha) = f_{\#}(\partial\alpha) = 0$ . Also,  $f_{\#}$  takes boundaries to boundaries since  $f_{\#}(\partial\beta) = \partial(f_{\#}\beta)$ . Hence  $f_{\#}$  induces a homomorphism  $f_* : H_n(X) \rightarrow H_n(Y)$ . An algebraic statement of what we have just proved is:

**Proposition 2.9.** *A chain map between chain complexes induces homomorphisms between the homology groups of the two complexes.*  $\square$

Two basic properties of induced homomorphisms which are important in spite of being rather trivial are:

- (i)  $(fg)_* = f_*g_*$  for a composed mapping  $X \xrightarrow{g} Y \xrightarrow{f} Z$ . This follows from associativity of compositions  $\Delta^n \xrightarrow{\sigma} X \xrightarrow{g} Y \xrightarrow{f} Z$ .
- (ii)  $\mathbb{1}_* = \mathbb{1}$  where  $\mathbb{1}$  denotes the identity map of a space or a group.

Less trivially, we have:

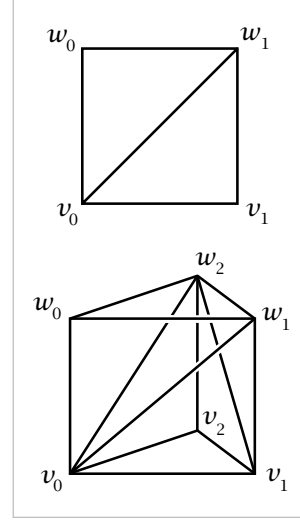
**Theorem 2.10.** *If two maps  $f, g : X \rightarrow Y$  are homotopic, then they induce the same homomorphism  $f_* = g_* : H_n(X) \rightarrow H_n(Y)$ .*

In view of the formal properties  $(fg)_* = f_*g_*$  and  $\mathbb{1}_* = \mathbb{1}$ , this immediately implies:

**Corollary 2.11.** *The maps  $f_* : H_n(X) \rightarrow H_n(Y)$  induced by a homotopy equivalence  $f : X \rightarrow Y$  are isomorphisms for all  $n$ .*  $\square$

For example, if  $X$  is contractible then  $\tilde{H}_n(X) = 0$  for all  $n$ .

**Proof of 2.10:** The essential ingredient is a procedure for subdividing the product  $\Delta^n \times I$  into  $(n+1)$ -simplices. The figure shows the cases  $n = 1, 2$ . In  $\Delta^n \times I$ , let  $\Delta^n \times \{0\} = [v_0, \dots, v_n]$  and  $\Delta^n \times \{1\} = [w_0, \dots, w_n]$ , where  $v_i$  and  $w_i$  have the same image under the projection  $\Delta^n \times I \rightarrow \Delta^n$ . The  $n$ -simplex  $[v_0, \dots, v_i, w_{i+1}, \dots, w_n]$  is the graph of the linear function  $\varphi_i: \Delta^n \rightarrow I$  defined in barycentric coordinates by  $\varphi_i(t_0, \dots, t_n) = t_{i+1} + \dots + t_n$  since the vertices of this simplex  $[v_0, \dots, v_i, w_{i+1}, \dots, w_n]$  are on the graph of  $\varphi_i$  and the simplex projects homeomorphically onto  $\Delta^n$  under the projection  $\Delta^n \times I \rightarrow \Delta^n$ . The graph of  $\varphi_i$  lies below the graph of  $\varphi_{i-1}$  since  $\varphi_i \leq \varphi_{i-1}$ , and the region between these two graphs is the simplex  $[v_0, \dots, v_i, w_i, \dots, w_n]$ , a true  $(n+1)$ -simplex since  $w_i$  is not on the graph of  $\varphi_i$  and hence is not in the  $n$ -simplex  $[v_0, \dots, v_i, w_{i+1}, \dots, w_n]$ . From the string of inequalities  $0 = \varphi_n \leq \varphi_{n-1} \leq \dots \leq \varphi_0 \leq \varphi_{-1} = 1$  we deduce that  $\Delta^n \times I$  is the union of the  $(n+1)$ -simplices  $[v_0, \dots, v_i, w_i, \dots, w_n]$ , each intersecting the next in an  $n$ -simplex face.



Given a homotopy  $F: X \times I \rightarrow Y$  from  $f$  to  $g$ , we can define *prism operators*  $P: C_n(X) \rightarrow C_{n+1}(Y)$  by

$$P(\sigma) = \sum_i (-1)^i F \circ (\sigma \times \mathbb{1}) | [v_0, \dots, v_i, w_i, \dots, w_n]$$

for  $\sigma: \Delta^n \rightarrow X$ , where  $F \circ (\sigma \times \mathbb{1})$  is the composition  $\Delta^n \times I \rightarrow X \times I \rightarrow Y$ . We will show that these prism operators satisfy the basic relation

$$\partial P = g_{\#} - f_{\#} - P\partial$$

Geometrically, the left side of this equation represents the boundary of the prism, and the three terms on the right side represent the top  $\Delta^n \times \{1\}$ , the bottom  $\Delta^n \times \{0\}$ , and the sides  $\partial \Delta^n \times I$  of the prism. To prove the relation we calculate

$$\begin{aligned} \partial P(\sigma) &= \sum_{j \leq i} (-1)^i (-1)^j F \circ (\sigma \times \mathbb{1}) | [v_0, \dots, \widehat{v}_j, \dots, v_i, w_i, \dots, w_n] \\ &\quad + \sum_{j \geq i} (-1)^i (-1)^{j+1} F \circ (\sigma \times \mathbb{1}) | [v_0, \dots, v_i, w_i, \dots, \widehat{w}_j, \dots, w_n] \end{aligned}$$

The terms with  $i = j$  in the two sums cancel except for  $F \circ (\sigma \times \mathbb{1}) | [\widehat{v}_0, w_0, \dots, w_n]$ , which is  $g \circ \sigma = g_{\#}(\sigma)$ , and  $-F \circ (\sigma \times \mathbb{1}) | [v_0, \dots, v_n, \widehat{w}_n]$ , which is  $-f \circ \sigma = -f_{\#}(\sigma)$ . The terms with  $i \neq j$  are exactly  $-P\partial(\sigma)$  since

$$\begin{aligned} P\partial(\sigma) &= \sum_{i < j} (-1)^i (-1)^j F \circ (\sigma \times \mathbb{1}) | [v_0, \dots, v_i, w_i, \dots, \widehat{w}_j, \dots, w_n] \\ &\quad + \sum_{i > j} (-1)^{i-1} (-1)^j F \circ (\sigma \times \mathbb{1}) | [v_0, \dots, \widehat{v}_j, \dots, v_i, w_i, \dots, w_n] \end{aligned}$$

Now we can finish the proof of the theorem. If  $\alpha \in C_n(X)$  is a cycle, then we have  $g_{\#}(\alpha) - f_{\#}(\alpha) = \partial P(\alpha) + P\partial(\alpha) = \partial P(\alpha)$  since  $\partial\alpha = 0$ . Thus  $g_{\#}(\alpha) - f_{\#}(\alpha)$  is a boundary, so  $g_{\#}(\alpha)$  and  $f_{\#}(\alpha)$  determine the same homology class, which means that  $g_*$  equals  $f_*$  on the homology class of  $\alpha$ .  $\square$

The relationship  $\partial P + P\partial = g_{\#} - f_{\#}$  is expressed by saying  $P$  is a **chain homotopy** between the chain maps  $f_{\#}$  and  $g_{\#}$ . We have just shown:

**Proposition 2.12.** *Chain-homotopic chain maps induce the same homomorphism on homology.*  $\square$

There are also induced homomorphisms  $f_* : \tilde{H}_n(X) \rightarrow \tilde{H}_n(Y)$  for reduced homology groups since  $f_{\#}\varepsilon = \varepsilon f_{\#}$ . The properties of induced homomorphisms we proved above hold equally well in the setting of reduced homology, with the same proofs.

## Exact Sequences and Excision

It would be nice if there was always a simple relationship between the homology groups of a space  $X$ , a subspace  $A$ , and the quotient space  $X/A$ . For then one could hope to understand the homology groups of spaces such as CW complexes that can be built inductively from successively more complicated subspaces. Perhaps the simplest possible relationship would be if  $H_n(X)$  contained  $H_n(A)$  as a subgroup and the quotient group  $H_n(X)/H_n(A)$  was isomorphic to  $H_n(X/A)$ . While this does hold in some cases, if it held in general then homology theory would collapse totally since every space  $X$  can be embedded as a subspace of a space with trivial homology groups, namely the cone  $CX = (X \times I)/(X \times \{0\})$ , which is contractible.

It turns out that this overly simple model does not have to be modified too much to get a relationship that is valid in fair generality. The novel feature of the actual relationship is that it involves the groups  $H_n(X)$ ,  $H_n(A)$ , and  $H_n(X/A)$  for all values of  $n$  simultaneously. In practice this is not as bad as it might sound, and in addition it has the pleasant side effect of sometimes allowing higher-dimensional homology groups to be computed in terms of lower-dimensional groups, which may already be known by induction for example.

In order to formulate the relationship we are looking for, we need an algebraic definition which is central to algebraic topology. A sequence of homomorphisms

$$\cdots \longrightarrow A_{n+1} \xrightarrow{\alpha_{n+1}} A_n \xrightarrow{\alpha_n} A_{n-1} \longrightarrow \cdots$$

is said to be **exact** if  $\text{Ker } \alpha_n = \text{Im } \alpha_{n+1}$  for each  $n$ . The inclusions  $\text{Im } \alpha_{n+1} \subset \text{Ker } \alpha_n$  are equivalent to  $\alpha_n \alpha_{n+1} = 0$ , so the sequence is a chain complex, and the opposite inclusions  $\text{Ker } \alpha_n \subset \text{Im } \alpha_{n+1}$  say that the homology groups of this chain complex are trivial.